

Source Characterization of The Hypergraphic Posets

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Abstract

For a hypergraph $\mathbb{H}$ on $[n]$, the hypergraphic poset $P_{\mathbb{H}}$ is the transitive closure of the oriented 1-skeleton of the hypergraphic polytope $\Delta_{\mathbb{H}}$.

In 2019, C. Benedetti, N. Bergeron, and J. Machacek established a correspondence between the transitive closure of the oriented 1-skeleton of $\Delta_{\mathbb{H}}$ and the flip graph on acyclic orientations of $\mathbb{H}$. More recently, N. Bergeron and V. Pilaud characterized $P_{\mathbb{H}}$ in terms of the sources of acyclic orientations for interval hypergraphs.

This work aims to extend this source characterization of $P_{\mathbb{H}}$ to arbitrary hypergraphs.

Hypergraphic Polytopes $\Delta_{\mathbb{H}}$

Definition: A *hypergraph* $\mathbb{H}$ on $[n]$ is a set of subsets of $[n] := \{1, \dots, n\}$. By convention, we always assume that singletons $\{i\}$ for $i \in [n]$ are included in $\mathbb{H}$.

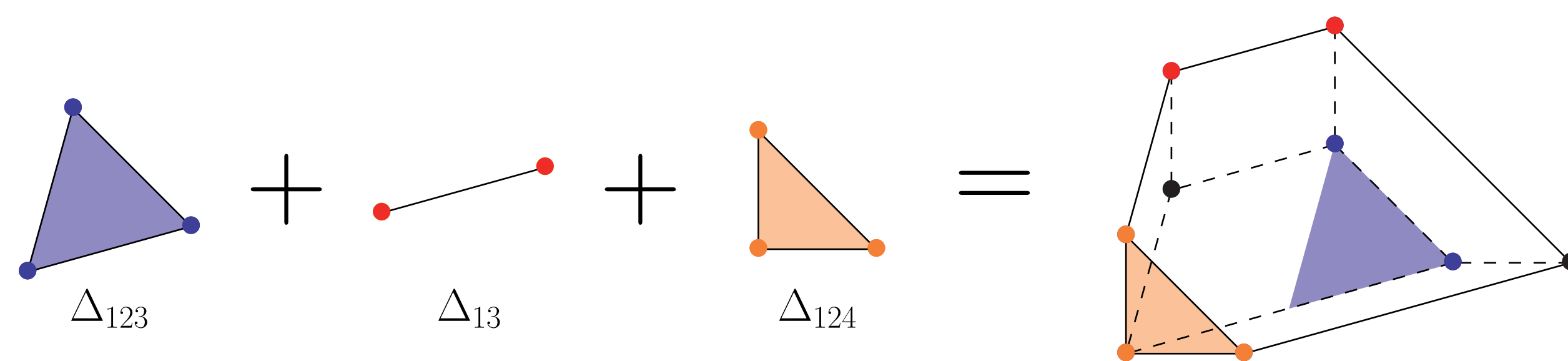
Notation: Let $e_i \in \mathbb{R}^n$ denote the standard basis vector in $\mathbb{R}^n$.

Definition: The *standard simplex* Δ_H is $\Delta_H = \text{conv}\{e_h | h \in H\}$.

Definition: The *hypergraphic polytope* $\Delta_{\mathbb{H}}$ is the Minkowski sum:

$$\Delta_{\mathbb{H}} = \sum_{H \in \mathbb{H}} \Delta_H.$$

Example: Hypergraphic polytope $\Delta_{\mathbb{H}}$ for the hypergraph $\mathbb{H} = \{123, 13, 124\}$.



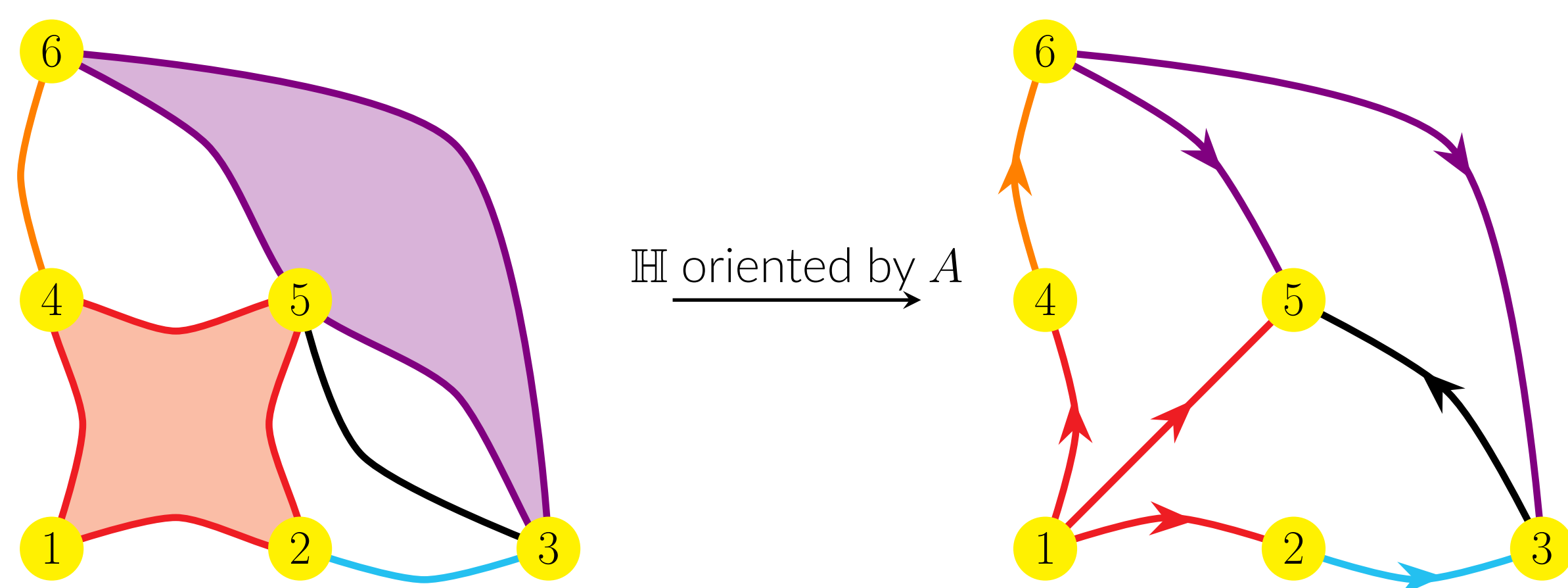
Orientation of a Hypergraph

Definition: An orientation of $\mathbb{H}$ is a map A from $\mathbb{H}$ to $[n]$ such that $A(H) \in H$ for all $H \in \mathbb{H}$.

Definition: The orientation A is *acyclic* if there is no $H_1, \dots, H_k$ with $k \geq 2$ such that $A(H_{i+1}) \in H_i \setminus \{A(H_i)\}$ for $i \in [k-1]$ and $A(H_1) \in H_k \setminus \{A(H_k)\}$.

Notation: We denote an orientation A by the source sequence $S_A = (A(H))_{H \in \mathbb{H}}$.

Example: Hypergraph $\mathbb{H} = \{1245, 23, 35, 46, 356\}$ oriented by $S_A = (1, 2, 3, 4, 6)$.

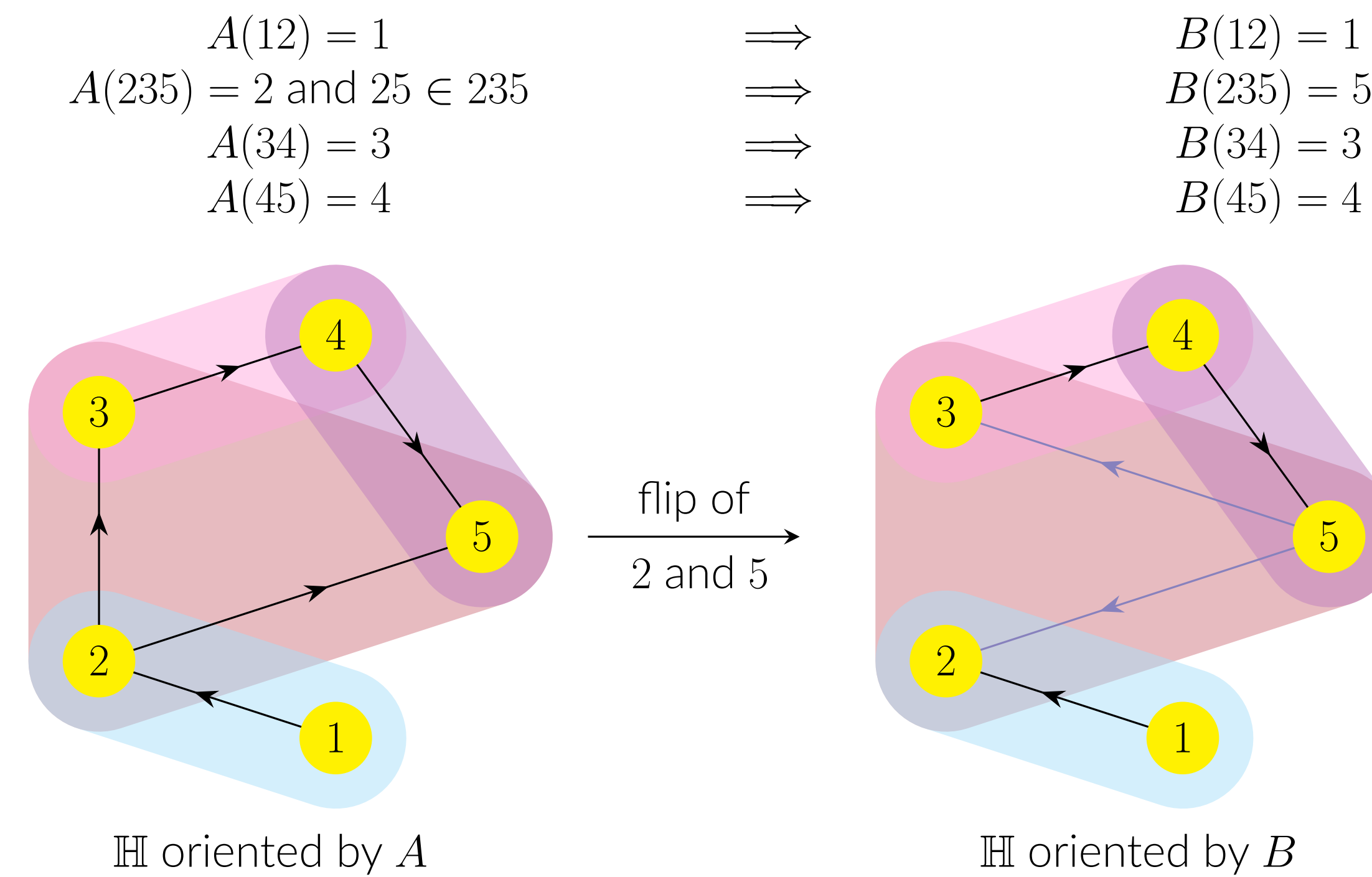


Flip of an Orientation

Definition: Two orientations $A \neq B$ of $\mathbb{H}$ are related by an *increasing flip* $A - ij \rightarrow B$ if there exist $1 \leq i < j \leq n$ such that for all $H \in \mathbb{H}$,

- if $A(H) \neq B(H)$ then $A(H) = i$ and $B(H) = j$, and
- if $\{i, j\} \subseteq H$, then $A(H) = i \iff B(H) = j$.

Example: Flip $A - 25 \rightarrow B$ on the hypergraph $H = \{12, 235, 34, 45\}$ between orientations $S_A = (1, 2, 3, 4)$ and $S_B = (1, 5, 3, 4)$.



Definition: The *flip graph* on acyclic orientations of a hypergraph $\mathbb{H}$ is the graph whose vertices are the acyclic orientations of $\mathbb{H}$, and where two orientations are connected by an edge if they differ by a single flip.

Surjection $\mathcal{O}$

Definition: Let $\mathcal{O}$ be a surjection described in [BBM19] from permutations $\mathfrak{S}_n$ to acyclic orientations of $\mathbb{H}$, where the acyclic orientation $\mathcal{O}_\pi$ of $\mathbb{H}$ is defined for all $H \in \mathbb{H}$ by

$$\mathcal{O}_\pi(H) := \pi(\min\{j | \pi(j) \in H\}).$$

Example: $\mathbb{H} = \{H_1 = \{1, 2, 3\}, H_2 = \{1, 2, 3, 4\}, H_3 = \{2, 3\}\}$

$$\begin{aligned} \mathcal{O}_{4132}(H_1) &= \frac{4 \ 1 \ 3 \ 2}{1} \\ \mathcal{O}_{4132}(H_2) &= 4 \\ \mathcal{O}_{4132}(H_3) &= 3 \end{aligned}$$

$$S_{\mathcal{O}_{4132}} = (1, 4, 3)$$

Proposition: Two distinct acyclic orientations A, B of $\mathbb{H}$ are related by the flip $A - ij \rightarrow B$ if and only if there are permutations π_A, π_B of $[n]$ such that $\pi_A = \pi_B(ij)$ and $\mathcal{O}_{\pi_A} = A$ and $\mathcal{O}_{\pi_B} = B$.

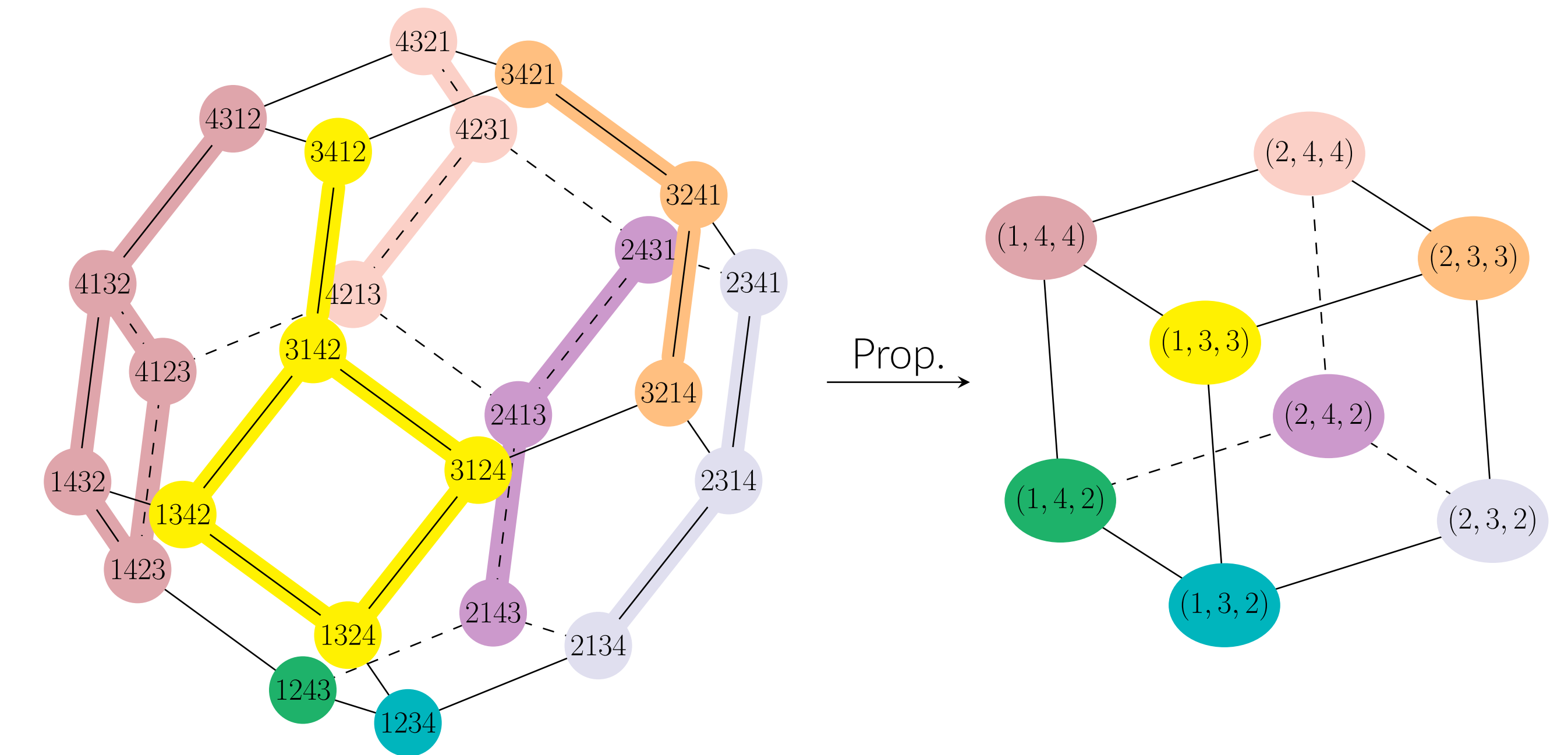
In other words, the 1-skeleton of $\Delta_{\mathbb{H}}$ is isomorphic to the graph obtained by contracting the fibers of $\mathcal{O}$ in the graph of the permutahedron.

Hypergraphic Posets $P_{\mathbb{H}}$

Geometric Definition: The *hypergraphic poset* $P_{\mathbb{H}}$ is the transitive closure of the 1-skeleton of $\Delta_{\mathbb{H}}$ oriented by $\omega := (n, n-1, \dots, 2, 1) - (1, 2, \dots, n-1, n) = (n-1, n-3, \dots, 3-n, 1-n)$.

Combinatorial Definition [BP24]: The *hypergraphic poset* $P_{\mathbb{H}}$ is the transitive closure of the increasing flip graph on acyclic orientations of $\mathbb{H}$.

Example: Hypergraphic polytopes $\Delta_{\mathbb{H}}$ and hypergraphic poset $P_{\mathbb{H}}$ according to the hypergraph $\mathbb{H} = \{12, 34, 234\}$.



Source Characterization

Theorem [GF25]: For any acyclic orientations A and B of a hypergraph $\mathbb{H}$

$$A \leq B \iff A(H) \leq B(H) \text{ for all } H \in \mathbb{H}.$$

Idea of Proof: ($\implies$) The forward implication of this theorem is given by the combinatorial definition of $P_{\mathbb{H}}$.

($\impliedby$) Let A and B be two distinct acyclic orientations of a hypergraph $\mathbb{H}$, such that $S_A < S_B$. To prove the backward implication of this theorem, we need to show the existence of a sequence of some k increasing flips

$$A - i_1 j_1 \rightarrow A_1 - i_2 j_2 \rightarrow A_2 - i_3 j_3 \rightarrow \dots \rightarrow A_{k-1} - i_k j_k \rightarrow A_k - i_k j_k \rightarrow B$$

such that

$$S_A < S_{A_1} < S_{A_2} < \dots < S_{A_k} < S_B.$$

To do so, we characterize cover relations in $P_{\mathbb{H}}$ using particular paths λ of orientations, called *coherent paths*.

To show the existence of a coherent flip $A - ij \rightarrow A'$ such that:

- $\sum_{H \in \mathbb{H}} (B(H) - A'(H)) < \sum_{H \in \mathbb{H}} (B(H) - A(H))$, and
- $S_A < S_{A'} \leq S_B$,

we algorithmically construct a path κ in A , called *source path* of a non-coherent flip $A - ij \rightarrow A'$.

References

- [BBM19] Carolina Benedetti, Nantel Bergeron, and John Machacek. Hypergraphic polytopes: combinatorial properties and antipode. *Journal of Combinatorics*, 10(3):515–544, 2019.
- [BP24] Nantel Bergeron and Vincent Pilaud. Interval hypergraphic lattices, 2024.